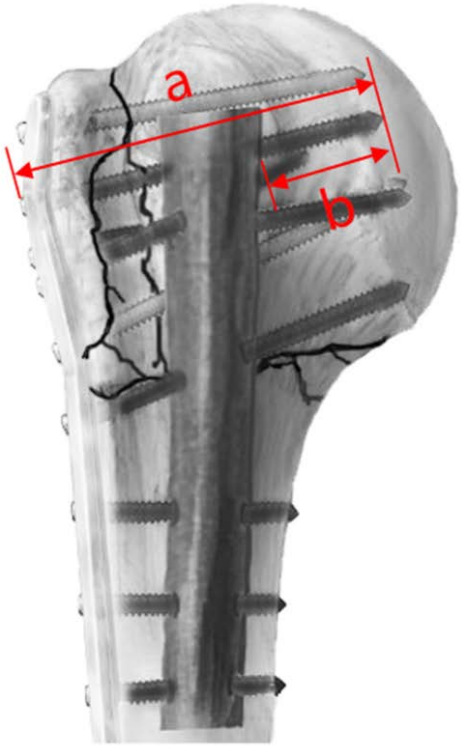
Additional file 5.



The fibular cortical segment and the traversed locking screws could be regard as a variation of an intramedullary nail. Along the lines of these screws, this “intramedullary nail” could provide support force to the medial column by the lever arm effect. Besides, the distances of the force arms of locking screws were also shortened (from a to b), which might reduce the risk of screw breakage.